

# Chapter 2

## Matrix Arithmetic

# Matrix Addition and Subtraction

- Addition and subtraction act *element-wise* on matrices.
- In order for the addition/subtraction ( $\mathbf{A} \pm \mathbf{B}$ ) to be possible, the two matrices  $\mathbf{A}$  and  $\mathbf{B}$  *must* have the same size. Then, we simply add/subtract the corresponding elements of each matrix to get the result:

$$(\mathbf{A} + \mathbf{B})_{ij} = A_{ij} + B_{ij}$$

$$(\mathbf{A} - \mathbf{B})_{ij} = A_{ij} - B_{ij}$$

- For example:

$$\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} + \begin{pmatrix} 4 & 5 \\ 6 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 7 & 5 \end{pmatrix}$$

## Example: Addition and Subtraction

a) Compute  $\mathbf{A} + \mathbf{B}$ , if possible:

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 4 & 5 & 6 \\ -1 & 0 & 4 \\ 3 & 4 & 3 \end{pmatrix}$$

*We can add the matrices because they have the same size.*

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 6 & 8 & 5 \\ 0 & -1 & 5 \\ 5 & 6 & 4 \end{pmatrix}$$

b) Compute  $\mathbf{A} - \mathbf{H}$ , if possible:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \quad \mathbf{H} = \begin{pmatrix} 6 & 5 & 10 \\ 0.1 & 0.5 & 0.9 \end{pmatrix}$$

*We cannot subtract these matrices because they don't have the same size.*

# Scalar Multiplication

$$\alpha \mathbf{A}$$

Multiplying a matrix by a scalar also works element-wise:

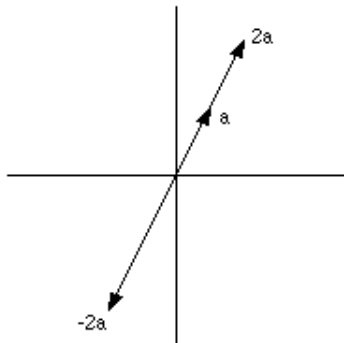
$$(\alpha \mathbf{A})_{ij} = \alpha A_{ij}$$

For example,

$$2 \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 2 & 4 \end{pmatrix}$$

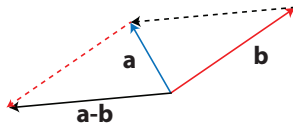
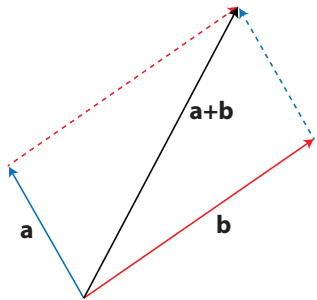
# Geometry of Scalar Multiplication

Scalar multiplication changes the length of a vector but not the overall direction (although a negative scalar will scale the vector in the opposite direction through the origin).



# Geometry of Vector Addition/Subtraction

Geometrically, vector addition is witnessed by placing the two vectors,  $\mathbf{a}$  and  $\mathbf{b}$ , *tail-to-head*. The result,  $\mathbf{a} + \mathbf{b}$ , is the vector from the open tail to the open head. When subtracting vectors as  $\mathbf{a} - \mathbf{b}$  we simply add  $-\mathbf{b}$  to  $\mathbf{a}$ . The vector  $-\mathbf{b}$  has the same length as  $\mathbf{b}$  but points in the opposite direction.



# Linear Combinations

A **linear combination** is constructed from a set of terms

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

by multiplying each term by a scalar constant and adding the result:

$$\mathbf{c} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \sum_{i=1}^n \alpha_i \mathbf{v}_i$$

The coefficients  $\alpha_i$  are scalar constants and the terms,  $\{\mathbf{v}_i\}$  can be scalars, vectors, or matrices.

Most often, we will consider linear combinations where the terms  $\{\mathbf{v}_i\}$  are vectors.

# Elementary Linear Combinations

The simplest linear combination might involve columns of the identity matrix (elementary vectors):

$$\begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

We can easily picture this linear combination as a “breakdown into parts” where the parts give directions along the 3 coordinate axis.

# Linear Combination of Matrices

Write the matrix  $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$  as a linear combination of the following matrices:

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

Solution:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} = 1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 3 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 4 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + 2 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

# Chapter 2

## Matrix Multiplication

- Matrix Multiplication is NOT done element-wise.
- Matrix Multiplication is NOT commutative
  - $\implies$  ORDER MATTERS!!  $\mathbf{AB} \neq \mathbf{BA}!!$

# A Notational Note

For the remainder of this course, unless otherwise specified, we will consider vectors to be columns rather than rows. This makes the notation more simple because if  $\mathbf{x}$  is a column vector,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

then we can automatically assume that  $\mathbf{x}^T$  is a row vector:

$$\mathbf{x}^T = (x_1 \quad x_2 \quad \dots \quad x_n) .$$

# The Inner Product (Dot Product)

The **inner product** of two vectors,  $\mathbf{x}$  and  $\mathbf{y}$ , written  $\mathbf{x}^T \mathbf{y}$ , is defined as the sum of the product of corresponding elements in  $\mathbf{x}$  and  $\mathbf{y}$ :

$$\mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

If we write this out for two vectors with 4 elements each, we'd have:

$$\mathbf{x}^T \mathbf{y} = (x_1 \quad x_2 \quad x_3 \quad x_4) \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = x_1 y_1 + x_2 y_2 + x_3 y_3 + x_4 y_4$$

*Note: The inner product between vectors is only possible when the two vectors have the same number of elements!*

# The Inner Product

The inner product defines how we multiply

ROW VECTOR  $\star$  COLUMN VECTOR

The product of two rows or two columns is not defined! We will see why shortly!

# Example - Inner Products

Let

$$\mathbf{x} = \begin{pmatrix} -1 \\ 2 \\ 4 \\ 0 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 3 \\ 5 \\ 1 \\ 7 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} -3 \\ -2 \\ 5 \\ 3 \\ -2 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} 2 \\ -1 \\ 3 \\ -3 \\ -2 \end{pmatrix}$$

If possible, compute the following inner products:

a.  $\mathbf{x}^T \mathbf{y}$

$$\begin{aligned} \mathbf{x}^T \mathbf{y} &= (-1 \ 2 \ 4 \ 0) \begin{pmatrix} 3 \\ 5 \\ 1 \\ 7 \end{pmatrix} \\ &= (-1)(3) + (2)(5) + (4)(1) + (0)(7) \\ &= -3 + 10 + 4 = \boxed{11} \end{aligned} \tag{1}$$

# Example - Inner Products

Let

$$\mathbf{x} = \begin{pmatrix} -1 \\ 2 \\ 4 \\ 0 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 3 \\ 5 \\ 1 \\ 7 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} -3 \\ -2 \\ 5 \\ 3 \\ -2 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} 2 \\ -1 \\ 3 \\ -3 \\ -2 \end{pmatrix}$$

If possible, compute the following inner products:

b.  $\mathbf{x}^T \mathbf{v}$

This is not possible because  $\mathbf{x}$  and  $\mathbf{v}$  do not have the same number of elements

c.  $\mathbf{v}^T \mathbf{u}$

$$\mathbf{v}^T \mathbf{u} = (-3 \quad -2 \quad 5 \quad 3 \quad -2) \begin{pmatrix} 2 \\ -1 \\ 3 \\ -3 \\ -2 \end{pmatrix} = \boxed{6}$$

It should be clear from the definition that for all vectors  $\mathbf{x}$  and  $\mathbf{y}$ ,

$$\mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x}.$$

Also, if we take the inner product of a vector with itself, the result is the sum of squared elements in that vector:

$$\mathbf{x}^T \mathbf{x} = \sum_{i=1}^n x_i^2 = x_1^2 + x_2^2 + \cdots + x_n^2.$$

# Check Your Understanding

Let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} \quad \mathbf{e} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \mathbf{p} = \begin{pmatrix} 0.5 \\ 0.1 \\ 0.2 \\ 0 \\ 0.2 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} 10 \\ 4 \\ 3 \\ 2 \\ 1 \end{pmatrix} \quad \mathbf{s} = \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}$$

If possible, compute the following inner products:

a.  $\mathbf{v}^T \mathbf{e}$

d.  $\mathbf{p}^T \mathbf{u}$

b.  $\mathbf{e}^T \mathbf{v}$

e.  $\mathbf{v}^T \mathbf{v}$

c.  $\mathbf{v}^T \mathbf{s}$

# Check Your Understanding - Solution

Let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} \quad \mathbf{e} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \mathbf{p} = \begin{pmatrix} 0.5 \\ 0.1 \\ 0.2 \\ 0 \\ 0.2 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} 10 \\ 4 \\ 3 \\ 2 \\ 1 \end{pmatrix} \quad \mathbf{s} = \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}$$

If possible, compute the following inner products:

a.  $\mathbf{v}^T \mathbf{e} = 15$

d.  $\mathbf{p}^T \mathbf{u} = 6.2$

b.  $\mathbf{e}^T \mathbf{v} = 15$

e.  $\mathbf{v}^T \mathbf{v} = 55$

c.  $\mathbf{v}^T \mathbf{s} = \text{not possible}$

## $AB$

- Matrix multiplication is just a series of inner products done simultaneously in one operation.
- As with vectors, the operation is not always possible.
- The order in which you multiply matrices makes a big difference!

# Matrix Multiplication - Definition

Let  $\mathbf{A}_{m \times n}$  and  $\mathbf{B}_{k \times p}$  be matrices. The matrix product  $\mathbf{AB}$  is possible if and only if  $n = k$ ; that is, when the number of columns in  $\mathbf{A}$  is the same as the number of rows in  $\mathbf{B}$ . If this condition holds, then the dimension of the product,  $\mathbf{AB}$  is  $m \times p$  and the  $(ij)$ -entry of the product  $\mathbf{AB}$  is the inner product of the  $i^{th}$  row of  $\mathbf{A}$  and the  $j^{th}$  column of  $\mathbf{B}$ :

$$(\mathbf{AB})_{ij} = \mathbf{A}_{i\star} \mathbf{B}_{\star j}$$

# Steps to Compute a Matrix Product

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 0 & -2 \\ 2 & -3 \end{pmatrix}$$

When we first get started with matrix multiplication, we often follow a few simple steps:

- 1) Write down the matrices and their dimensions. Make sure the “inside” dimensions match - those corresponding to the columns of the first matrix and the rows of the second matrix:

$$\begin{matrix} \mathbf{A} & \mathbf{B} \\ (3 \times 2) & (2 \times 2) \end{matrix}$$

If these dimensions match, then we can multiply the matrices. If they don't, multiplication is not possible.

# Steps to Compute a Matrix Product

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 0 & -2 \\ 2 & -3 \end{pmatrix}$$

- 2) Now, look at the “outer” dimensions - this will tell you the size of the resulting matrix.

$$\begin{matrix} \mathbf{A} & \mathbf{B} \\ (3 \times 2) & (2 \times 2) \end{matrix}$$

So the product  $\mathbf{AB}$  is a  $3 \times 2$  matrix.

# Steps to Compute a Matrix Product

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 0 & -2 \\ 2 & -3 \end{pmatrix} \quad \mathbf{AB} = \begin{pmatrix} (\mathbf{AB})_{11} & (\mathbf{AB})_{12} \\ (\mathbf{AB})_{21} & (\mathbf{AB})_{22} \\ (\mathbf{AB})_{31} & (\mathbf{AB})_{32} \end{pmatrix}$$

- 3) Finally, we compute the product of the matrices by multiplying each row of  $\mathbf{A}$  by each column of  $\mathbf{B}$  using inner products.

$$(\mathbf{AB})_{11} = \mathbf{A}_{1\star} \mathbf{B}_{\star 1}$$

$$(\mathbf{AB})_{ij} = \mathbf{A}_{i\star} \mathbf{B}_{\star j}$$

# Steps to Compute a Matrix Product

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 0 & -2 \\ 2 & -3 \end{pmatrix}$$

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} (2)(0) + (3)(2) & (2)(-2) + (3)(-3) \\ (-1)(0) + (4)(2) & (-1)(-2) + (4)(-3) \\ (5)(0) + (1)(2) & (5)(-2) + (1)(-3) \end{pmatrix} \\ &= \begin{pmatrix} 6 & -13 \\ 8 & -10 \\ 2 & -13 \end{pmatrix} \end{aligned} \tag{2}$$

# Check Your Understanding

Suppose we have

$$\mathbf{A}_{4 \times 6} \quad \mathbf{B}_{5 \times 5} \quad \mathbf{M}_{5 \times 4} \quad \mathbf{P}_{6 \times 5}$$

Circle the matrix products that are possible to compute and write the dimension of the result.

$$\mathbf{AM} \quad \mathbf{MA} \quad \mathbf{BM} \quad \mathbf{MB} \quad \mathbf{PA} \quad \mathbf{PM} \quad \mathbf{AP} \quad \mathbf{A}^T \mathbf{P} \quad \mathbf{M}^T \mathbf{B}$$

# Check Your Understanding - Solution

Suppose we have

$$\mathbf{A}_{4 \times 6} \quad \mathbf{B}_{5 \times 5} \quad \mathbf{M}_{5 \times 4} \quad \mathbf{P}_{6 \times 5}$$

Circle the matrix products that are possible to compute and write the dimension of the result.

$$\mathbf{AM} \quad \textcircled{\mathbf{MA}}_{5 \times 6} \quad \textcircled{\mathbf{BM}}_{5 \times 4} \quad \mathbf{MB} \quad \mathbf{PA} \quad \textcircled{\mathbf{PM}}_{6 \times 4} \quad \textcircled{\mathbf{AP}}_{4 \times 5} \quad \mathbf{A}^T \mathbf{P} \quad \textcircled{\mathbf{M}^T \mathbf{B}}_{4 \times 5}$$

# Check Your Understanding

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \mathbf{M} = \begin{pmatrix} -2 & 1 & -1 & 2 & -2 \\ 1 & -2 & 0 & -1 & 2 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

Determine the following matrix products, if possible.

a.  $\mathbf{AC}$

b.  $\mathbf{AM}$

c.  $\mathbf{AC}^T$

# Check Your Understanding - Solution

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \mathbf{M} = \begin{pmatrix} -2 & 1 & -1 & 2 & -2 \\ 1 & -2 & 0 & -1 & 2 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

Determine the following matrix products, if possible.

a.  $\mathbf{AC} = \begin{pmatrix} 0 & -1 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$

b.  $\mathbf{AM} = \begin{pmatrix} -1 & -1 & -1 & 1 & 0 \\ 1 & -2 & 0 & -1 & 2 \end{pmatrix}$

c.  $\mathbf{AC}^T = \text{not possible}$

One very important thing to keep in mind is that

*matrix multiplication is **not** commutative!*

As we see from the previous exercises, it's quite common to be able to compute a product  $\mathbf{AB}$  where the reverse product,  $\mathbf{BA}$  is not even possible to compute. Even if both products are possible it is almost *never* the case that  $\mathbf{AB}$  equals  $\mathbf{BA}$

# Multiplication by a Diagonal Matrix

$$\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\mathbf{DA} = \begin{pmatrix} 2 & 4 & 6 \\ 3 & 3 & 6 \\ -4 & -2 & -6 \end{pmatrix}$$

In doing this multiplication, we see that the effect of multiplying the matrix  $\mathbf{A}$  by a diagonal matrix on the left is that the rows of the matrix  $\mathbf{A}$  are simply scaled by the entries in the diagonal matrix.

# Matrix-Vector Product

- A matrix-vector product works exactly the same way as matrix multiplication.
- Again we must match the dimensions to make sure they line up correctly.
- Suppose we have  $\mathbf{A}_{m \times n}$ . Then we can multiply a row vector  $\mathbf{v}_{1 \times m}^T$  on the left:

$$\mathbf{v}^T \mathbf{A} \text{ works because } \underset{(1 \times m)}{\mathbf{v}^T} \underset{(m \times n)}{\mathbf{A}}$$

$\implies$  The result will be a  $1 \times n$  row vector.

- Or, we can multiply an  $n \times 1$  column vector  $\mathbf{x}$  on the right:

$$\mathbf{A} \mathbf{x} \text{ works because } \underset{(m \times n)}{\mathbf{A}} \underset{(n \times 1)}{\mathbf{x}}$$

$\implies$  The result will be a  $m \times 1$  column vector.

# Matrix-Vector Product

- Matrix-vector multiplication works the same way as matrix multiplication: we simply multiply rows by columns until we've completed the answer.
- As we will see in the near future, there are many ways to view matrix multiplication.
- Multiplying a matrix by a vector amounts to taking a linear combination of its rows or columns.

# Matrix-Vector Products

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad \mathbf{q} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

Determine whether the following matrix-vector products are possible. When possible, compute the product.

a.  $\mathbf{Aq}$

Not Possible: Inner dimensions do not match  $\mathbf{A}_{(3 \times 2)} \mathbf{q}_{(3 \times 1)}$

b.  $\mathbf{Av}$

$$\begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2(3) + 3(2) \\ -1(3) + 4(2) \\ 5(3) + 1(2) \end{pmatrix} = \begin{pmatrix} 12 \\ 5 \\ 17 \end{pmatrix}$$

# Linear Combination of Columns

$$\begin{aligned}\mathbf{Av} &= \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2(\textcolor{red}{3}) + 3(\textcolor{blue}{2}) \\ -1(\textcolor{red}{3}) + 4(\textcolor{blue}{2}) \\ 5(\textcolor{red}{3}) + 1(\textcolor{blue}{2}) \end{pmatrix} \\ &= \textcolor{red}{3} \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} + \textcolor{blue}{2} \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 12 \\ 5 \\ 17 \end{pmatrix} \quad (3)\end{aligned}$$

# Why Does this Viewpoint Matter?

- Suppose I had a matrix of data, containing columns for *height*, *weight*, and *length*.
- If I wanted to make a new variable, that was equal to

$$0.5 \star \textit{height} + 2 \star \textit{weight} + 0.75 \star \textit{length}$$

All I would have to do is multiply my data matrix by the vector

$$\mathbf{v} = \begin{pmatrix} 0.5 \\ 2 \\ 0.75 \end{pmatrix}$$

# Vector Outer Product

There is one last type of product to consider: A column vector times a row vector.

- Regardless of the vectors' size, this will always be possible.
- The dimensions of the outcome?

$$\underset{(m \times \textcolor{red}{1})}{\mathbf{x}} \underset{(\textcolor{red}{1} \times n)}{\mathbf{y}^T} = \mathbf{M}_{m \times n}$$

$\implies$  The result is a matrix!

- Treat this product the same way we treat any matrix product, by multiplying row  $\times$  column until we've run out of rows and columns.

# Vector Outer Product - Example

Let  $\mathbf{x} = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix}$  and  $\mathbf{y} = \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}$ . Then,

$$\mathbf{xy}^T = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} \begin{pmatrix} 1 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 15 & 9 \\ 4 & 20 & 12 \\ -2 & -10 & -6 \end{pmatrix}$$

As you can see by performing this calculation, a vector outer product will *always* produce a matrix whose rows are multiples of each other!

# Chapter 2

## Special Cases of Matrix Multiplication

# The Identity Matrix

- The Identity Matrix, **I**, is to matrices as the number '1' is to scalars.
- It is the **multiplicative identity**.
- For any matrix (or vector) **A**, multiplying **A** by the identity matrix on either side does not change **A**:

$$\mathbf{AI} = \mathbf{A}$$

$$\mathbf{IA} = \mathbf{A}$$

- This is easy to verify since the identity matrix is simply a diagonal matrix with ones on the diagonal.

# The Matrix Inverse

For certain square matrices,  $\mathbf{A}$ , an inverse matrix, written  $\mathbf{A}^{-1}$ , exists such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

- A MATRIX MUST BE SQUARE TO HAVE AN INVERSE
- NOT ALL SQUARE MATRICES HAVE AN INVERSE
- Only full-rank, square matrices are invertible. We will talk about rank soon.

For now, understand that the inverse matrix serves like the multiplicative inverse in scalar algebra:

$$(2)(2^{-1}) = (2)\left(\frac{1}{2}\right) = 1$$

Multiplying a matrix by its inverse (if it exists) yields the multiplicative identity,  $\mathbf{I}$ .

A square matrix which has an inverse is equivalently called:

- **Non-singular**
- **Invertible**

# Don't cancel that!!

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

$$\mathbf{A}\cancel{\mathbf{x}} = \lambda\cancel{\mathbf{x}}$$

??????????????

$$\mathbf{A}_{m \times n} \mathbf{x}_{n \times 1} = \lambda_{1 \times 1} \mathbf{x}_{m \times 1}$$

$\mathbf{A}$  is a matrix and  $\lambda$  is a scalar!  $\mathbf{A} = \lambda$  is **nonsense**!

# Cancellation implies multiplication by an inverse

Canceling numbers in algebra:

$$2x = 2y$$

$$\cancel{2}x = \cancel{2}y$$

$$\frac{1}{\cancel{2}}\cancel{2}x = \frac{1}{\cancel{2}}\cancel{2}y$$

$$1x = 1y$$

Canceling matrices in *linear* algebra:

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

$$\mathbf{A}^{-1}\mathbf{A}\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

$$\mathbf{I}\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

So inverses are great - WHEN THEY EXIST!